

Fraud Pattern Analysis: Executive Insight Report

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Executive Summary: Coordinated Account Takeover and Emulator Fraud

- The behavioral and network investigation reveals a highly sophisticated and coordinated fraud operation targeting user accounts across multiple banks. The observed patterns including rapid rotation of many IP addresses, a singular device fingerprint being cloned across many unique devices, and the strong geographic concentration of activity are conclusive evidence of automated Account Takeover (ATO) campaigns run from virtual environments (emulators) and infrastructure designed for evasion.
- The critical link is a shared, abnormal network and device profile between the previously confirmed Compromised Identity (Bank8) and the Compromised Device (Bank4), confirming that both entities belong to the same attacker infrastructure.

Analytical Methodology Highlight

- To move beyond basic frequency analysis, the investigation utilized specialized Python modules. The GeoIP library was essential for enriching the raw data, allowing the analysis to accurately generate critical network and geographic risk factors (Country, Provider, and ASN) that link the attacks to Russia. Furthermore, Graph Network Libraries were used for visually mapping the non-obvious links between compromised Identities, Devices, and IPs, proving the unified infrastructure, and Advanced Clustering Algorithms were deployed to automatically group high volume entities that exhibited non-human, repetitive patterns

Key Findings and Technical Indicators

The initial analysis of the two known fraud anchors established a clear link:

- Shared Network: Both the Compromised Identity and the Compromised Device utilized IP addresses routed through Russia and shared the same Autonomous System Number (ASN 8359.0).
- Compromised Identity (Bank8): This single user identity exhibits extreme behavior, connecting from 70 unique IP addresses and across 4 distinct geographic locations (all within Russia, with Moscow accounting for 54 logs). This behavior is characteristic of rapid IP cycling via proxies or VPNs to evade velocity checks.
- Compromised Device (Bank4): This device was used to access 2 different accounts (multiple identities) and logged a “No data” device fingerprint in all 3 records. The use of “No data” fingerprints and multi-identity access from a singular device is a strong Indicator of device spoofing/emulation.
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1. Evidence of Large-Scale Automation and Device Emulation

- The data demonstrates the use of high-volume, automated tooling to bypass device-based security measures:
- Cloned Fingerprint for Evasion: A single, consistent device fingerprint was found to be used by 113 different unique device IDs. This strongly suggests a “gold standard” fingerprint was successfully extracted and cloned within a controlled emulator environment to execute mass attacks.
- High-Frequency Abuse: A small set of core entities are driving the majority of suspicious activities. The top device ID was logged 498 times, and the top identity was logged 496 times.
- Shared IP/Identity Abuse: Out of the top 10 most frequent IP addresses, 8 of them were associated with an abnormally high number of unique accounts, ranging from 2 to 39 unique identities and 2 to 46 unique devices. The top IP (185.97.201.65) appeared 50 times.

3. Network and Geographic Concentration

- The activity is concentrated on specific network blocks, facilitating targeted blocking:
- Geographic Origin: The primary source of anomalous traffic is Russia, which accounts for a total of 1269 records in the dataset.
- Network Providers: The top three Autonomous System Numbers (ASNs) are major enablers of this traffic: T2 Mobile LLC (393), MTS PJSC (277), and PVimpelCom (246).

Recommendations

The consistent, shared, and highly abnormal patterns necessitate an immediate, infrastructure focused response:

- Immediate Entity Blockade: Immediately place all accounts associated with the top 10 most frequent identities and devices on high-risk watches or block them entirely including the compromised identity ([07f265568f1abc1da791e85287df2047b3af9](#)).
- Emulator Detection Enhancement: Implement a global rule to flag or block any new device attempting to use the cloned, high-frequency device fingerprint (which was found across 113 devices) as a near-certain automated fraud attempt.
- Geographic and Network Risk Scoring: Immediately increase the risk score for all transactions originating from the top facilitating ASNs (e.g., T2 Mobile LLC, ASN 8359.0) and the concentrated Russian IP ranges.
- Behavioral Anomaly Rule Creation: Deploy a rule to detect the specific fraud signature identified: High Unique IP Count (e.g., >10 unique IPs in a short period) coupled with a Single/No Device Fingerprint (as seen with the compromised entities). This pattern is the hallmark of scripted account takeover